

ANJALI DESAI

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Summary

Anjali Desai is a 12th grade student at National Hill View Public School, Bangalore, with a passion for AI research, competitive mathematics, and community-driven technology. CBSE student with a 97% academic average across 9th–11th grade and a 1560 SAT score. First author on a conference paper submitted to IIM Bangalore on Human-AI synergy in school education, awarded Outstanding Research Intern; also published a breast cancer whitepaper, a breast cancer subtype classification preprint, and has an ACM CSUR survey paper in progress. USACO Gold qualifier (top 5% globally), AMO Global Rank 29 (2024) and Global Rank 9 (2025), Singapore Math Global Gold Medallist, and UCO national topper. Vice-Director of Digital Innovation in Student Council, Chief Editor of the school's first quarterly magazine, Treasurer of AETHRA (\$17,000+ raised across 6 global events), and Tech Lead of Shakti Society.

Education

National Hill View Public School, Bangalore

Graduating March 2027

CBSE — Physics, Chemistry, Mathematics, Computer Science

- Academic Average: 97% (9th), 97% (10th), 97% (11th)
- SAT: 1560 (Math 780 / English 780)
- AP Scores (May 2026): Computer Science A (5), Physics C Mechanics (5), Statistics (5), Calculus BC (5)

Research & Publications

Beyond Artificial Intelligence: Designing Human-AI Synergy Model to Achieve Sustainable, Equitable, and Global School Education in India 2025

- Conference paper submitted to IIM Bangalore; first author, researched under a professor.
- Proposes a framework for integrating human-AI collaboration to advance equitable and sustainable school education in India.
- Awarded Outstanding Research Intern.

AI-Augmented Multimodal Breast Cancer Risk Prediction — Whitepaper Published 2025

- Proposes an AI-driven multimodal framework for early breast cancer risk prediction using polygenic scores, lifestyle data, and temporal EMR modeling.
- Published on [OneYoungIndia](https://www.oneyoungindia.com).

Multimodal Deep Learning for Breast Cancer Subtype Classification Jul – Oct 2024

- Integrated histopathology images and biomarker data to classify four breast cancer subtypes; 87% improvement in macro-F1 score (0.678 vs 0.363 baseline) using ResNet18/50 and DINO ViT.
- Published preprint [here](#).

Intersectional Computing and Equitable Artificial Intelligence for Sustainable Development: A Survey In Progress

- Survey paper for ACM Computing Surveys (CSUR), co-authored under a professor.
- Surveys the intersection of equitable AI design, intersectional computing, and sustainable development goals.

Projects

BuzzGuard — Offline Mosquito Detection App 2025

- Built an offline phone app that detects disease-carrying mosquitoes (Aedes aegypti, Anopheles, Culex) by analysing wingbeat audio via an on-device TensorFlow.js CNN — no internet or hardware required beyond a phone.
- Fuses acoustic detections with stagnant-water reports and rainfall data into a ward-level risk score, generating concrete task prioritisation for health workers (e.g. 'Act now — send a team for source reduction').
- Trained on ~279k lab recordings (Wingbeats dataset); achieved 75.6% test accuracy and 100% background recall. Live demo: [buzzguard app](#).

Bangalore AQI Forecasting — GRU Deep Learning Model 2025

- Developed a stacked GRU neural network for 7-day air quality forecasting across Bangalore, incorporating 7 pollutant parameters (PM2.5, PM10, NO2, SO2, CO, O3, AQI) with a 30-day sliding window.
- Achieved R^2 of 0.70 and RMSE of 13.41; model converged in 40 epochs with early stopping and adaptive learning rate scheduling.

Achievements & Awards

USACO (USA Computing Olympiad)	2025
• Qualified for Gold Division — top 5% of participants globally.	
American Mathematics Olympiad (AMO)	2024 & 2025
• 2024: Global Rank 29, Country Rank 6 — Global Bronze Medal. • 2025: Global Rank 9, Country Rank 3.	
Singapore Math Global Final	2024
• Country Rank 1, Global Rank 14 — Gold Medal.	
Unified Cyber Olympiad (UCO)	Multiple Years
• National 1st and National 2nd across consecutive years in computer science and cybersecurity.	
Global Essay and Music Awards (GEMA)	2023
• National winner in short story competition; winning story published as prize.	

Leadership & Extracurriculars

Vice-Director of Digital Innovation, Student Council — National Hill View Public School	May 2025 – Present
• Spearheaded school-wide AI integration club with workshops, mentoring, and competitions for students. • Organised a Hackathon workshop in partnership with Microsoft at the Microsoft campus for 11th grade students. • Manage and contribute to school social media presence, creating content and driving digital engagement.	
Chief Editor, School Magazine Editorial Board — National Hill View Public School	Sep 2025 – Present
• Initiated and led the creation of the school's first ever quarterly magazine. • Lead editorial direction, content curation, and publication across each issue.	
Treasurer — AETHRA (Student-Led Tech Organisation)	Jun 2025 – Present
• Raised \$17,000+ in prizes from company sponsors across 6 global online events. • Host a wide variety of events including hackathons, ideathons, pitchathons, vibeathons, brandathons, and gameathons.	
Tech Lead — Shakti Society (Women's Empowerment Initiative)	Jun 2025 – Present
• Built and maintain full-stack website with event management, resource library, and social media integration. • Lead technology strategy and digital presence for all online programmes and outreach.	

Courses & Certifications

- **Brown University** Pre-College AI, ML & Data Science — A+ (98%); led team of 4 in stock market prediction model (86%+ directional accuracy).
- **IIT Madras** AI Course (8 weeks); **AGSD** AI Bootcamp (12 weeks) with mentorship on breast cancer classification project.
- **PCEP** — Python Certified Entry-Level Programmer; **Java Specialization** (5 courses, Codio); **C++ Specialization** (5 courses, Codio); Python (Google).

Volunteering

Digital Literacy Volunteer — CRY Foundation, Bangalore	2022 – Present
• CRY (Child Rights and You) works to protect the rights of underprivileged children across India; contributed by delivering sessions on cybersecurity awareness and digital literacy to students with limited access to technology. • 60+ hours/semester over nearly 4 years; led volunteer teams during community events including the Walk to EmpowHER walkathon.	

Technical Skills

Languages: Python, Java, C++, JavaScript, HTML/CSS

AI/ML: PyTorch, TensorFlow, Keras, scikit-learn, Pandas, NumPy, Matplotlib

Tools: GitHub, Jupyter Notebook, Google Colab, Streamlit, VS Code

Creative Pursuits

- **Creative Writing:** [Blog](#) with nearly 70 original short stories; Kindle e-book; won GEMA national short story competition and multiple inter-school awards.
- **Music:** Trinity College London Guitar Grades 1–4 (**Outstanding**); Grade 5 in progress; active stage performer and [YouTube creator](#).